Expt: 11a

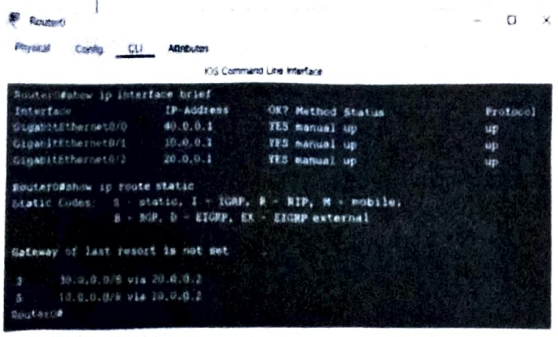
Simulate Static routing configuration using cisco Packet Tracer.

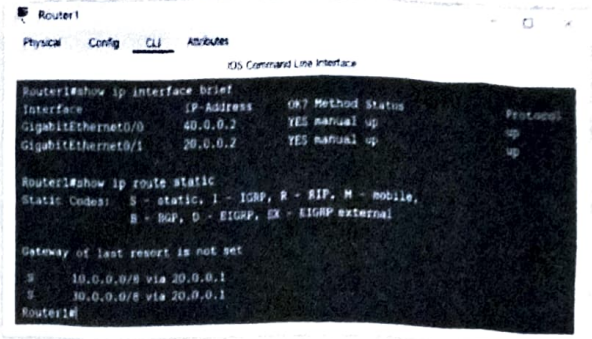
Aim:

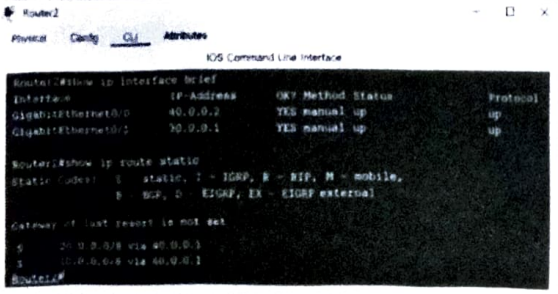
To simulate a static routing configuration using cisco packet tracer.

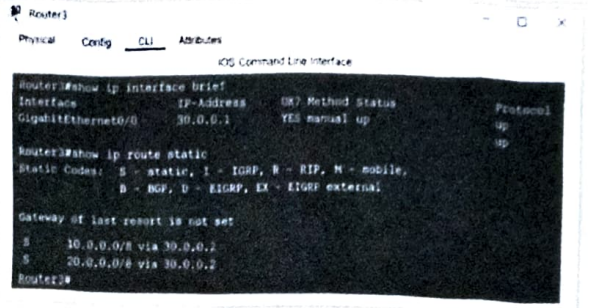
Procedure:

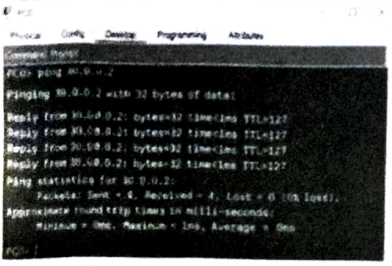
- Open Cisco Packet Tracer and create the given network topology.
- Connect 3 routers and PC's using suitable cable
- configure IP address for routers and PC's
- Enable all router interface using no shutdown
- Configure static routes on Router 0, router1, router2 using ip mobile
- Set main and backup router using different administrative distance values.
- Test connectivity
- Use tracert to verify the routing path.
- Disconnect the main link and verify that the backup router works.
- Thus a static routing is successfully configured and verified

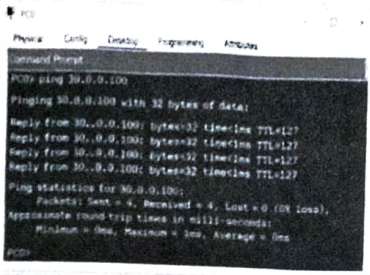

Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router0#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 40.0.0.1 YES manual up up
GigabitEthernet0/1 30.0.0.1 YES manual up up
GigabitEthernet0/2 20.0.0.1 YES manual up up
Router0#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 30.0.0.0/8 via 20.0.0.2
S 10.0.0.0/8 via 20.0.0.2
Router0#

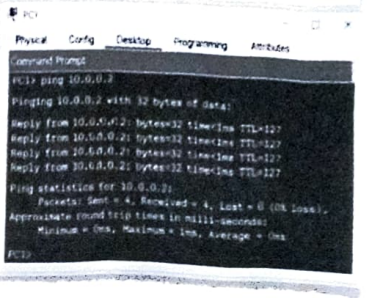

Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 40.0.0.2 YES manual up up
GigabitEthernet0/1 20.0.0.2 YES manual up up
Router1#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 10.0.0.0/8 via 20.0.0.1
S 30.0.0.0/8 via 20.0.0.1
Router1#


Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router2#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 40.0.0.2 YES manual up up
GigabitEthernet0/1 30.0.0.1 YES manual up up
Router2#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 20.0.0.0/8 via 40.0.0.1
S 10.0.0.0/8 via 40.0.0.1
Router2#


Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router3#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 30.0.0.1 YES manual up up
Router3#show ip route static
Static Codes: S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
Gateway of last resort is not set
S 10.0.0.0/8 via 30.0.0.2
S 20.0.0.0/8 via 30.0.0.2
Router3#


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
PC0> ping 30.0.0.2
Pinging 30.0.0.2 with 32 bytes of data:
Reply from 30.0.0.2: bytes=32 time<1ms TTL=127
Reply from 30.0.0.2: bytes=32 time<1ms TTL=127
Reply from 30.0.0.2: bytes=32 time<1ms TTL=127
Reply from 30.0.0.2: bytes=32 time<1ms TTL=127
Ping statistics for 30.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC0>


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
PC0> ping 30.0.0.100
Pinging 30.0.0.100 with 32 bytes of data:
Reply from 30.0.0.100: bytes=32 time<1ms TTL=127
Reply from 30.0.0.100: bytes=32 time<1ms TTL=127
Reply from 30.0.0.100: bytes=32 time<1ms TTL=127
Reply from 30.0.0.100: bytes=32 time<1ms TTL=127
Ping statistics for 30.0.0.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC0>


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
PC1> ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time<1ms TTL=127
Reply from 10.0.0.2: bytes=32 time<1ms TTL=127
Reply from 10.0.0.2: bytes=32 time<1ms TTL=127
Reply from 10.0.0.2: bytes=32 time<1ms TTL=127
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC1>

Expt No: 11 b

Simulate RIP using cisco packet Tracer

Aim

To Simulate RIP using (3 routers and 2 PC's) cisco packet tracer.

Procedure:

- create the topology with 3 routers and 2PC's
- Connect the device using suitable cables
- Assign the given IP addresses to both pc's.
- Configure fast Ethernet and serial interface of all routers
- Enable interface using no shutdown.
- Set clock rate of 4000 and bandwidth 64 on DE serial interface
- Enable RIP on all routers using router RIP
- Add directly connected network using the network command.
- Test connectivity using ping 20.0.0.2
- Use tracer to 20.0.0.2 to verify the selected route.
- Disconnect the direct router 0 - router2 link and use tracert again to verify RIP selects the alternate route.

PC0 – Command Prompt (ping)

```
C:\> ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\> tracert 20.0.0.2
Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1      1 ms      1 ms      1 ms   10.0.0.1
  2      2 ms      2 ms      2 ms   192.168.1.253
  3      3 ms      3 ms      3 ms   20.0.0.2

Trace complete.

C:\>
```

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Result:

Thus static routing configuration & RIP is simulated using Cisco packet Tracer.